

MAUNA LOA 5 NNE, HI, USA

WMO: 912950

Lat: 19.5354N

Lon: 155.5760W

Elev: 11179

StdP: 9.65

Time Zone: -10.00 (W10)

Period: 05-19

WBAN: 21514

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	30.5	32.1	-43.6	0.7	47.2	-36.6	1.0	47.3	29.8	40.8	26.4	42.9	8.4	N/A	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	14.8	61.8	37.9	60.1	37.9	58.8	38.0	48.1	52.7	47.2	51.9	46.3	51.4	12.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
46.2	70.7	49.8	45.0	67.7	49.2	43.9	64.8	49.0	23.2	52.9	22.6	51.8	22.1	51.5	51.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	20.0	17.6	26.9	65.5	2.7	1.1	25.0	66.3	23.4	66.9	21.9	67.5	20.0	68.3		
(5)			17.9	50.0	1.6	0.9	16.7	50.6	15.7	51.1	14.8	51.6	13.6	52.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.8	43.4	41.8	42.6	44.6	46.7	49.1	48.8	48.7	48.2	47.5	44.7	43.5
	DBStd	4.47	4.61	4.45	3.94	3.80	3.81	3.41	3.39	3.03	2.88	2.93	3.46	3.96
	HDD50	1650	209	232	231	167	114	56	63	61	68	88	159	202
	HDD65	7000	671	651	694	613	566	477	502	504	504	544	608	666
	CDD50	125	4	1	1	3	13	29	26	22	15	9	1	1
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.4	8.7	8.8	8.9	8.6	8.4	8.9	8.7	8.3	7.1	7.6	7.6	9.0
Precipitation	PrecAvg	30.6	3.6	2.2	3.1	1.9	1.8	1.4	2.3	2.0	2.2	2.0	2.8	2.6
	PrecMax	53.8	14.5	6.1	8.0	8.0	7.4	4.5	6.1	3.8	5.9	4.7	11.6	6.9
	PrecMin	13.2	0.2	0.2	0.7	0.8	0.4	0.2	0.5	0.6	0.2	0.5	0.2	0.0
	PrecStd	10.2	3.3	1.7	2.4	1.5	1.6	0.9	1.5	0.8	1.5	1.2	2.7	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.8	58.1	58.7	59.7	62.5	63.3	63.9	63.1	61.8	61.2	57.9	58.6
		MCWB	35.8	34.2	36.6	36.4	38.1	39.1	37.6	38.4	37.5	36.6	35.9	34.2
	2%	DB	56.4	54.8	55.6	57.1	59.2	60.7	60.6	60.6	59.1	58.5	55.9	55.3
		MCWB	35.3	34.9	36.6	36.1	38.7	38.6	38.2	38.5	39.0	37.5	36.1	34.1
	5%	DB	54.1	52.3	53.3	54.9	56.9	58.9	58.8	58.5	57.0	56.4	54.0	53.2
		MCWB	34.4	34.7	36.2	37.2	39.1	38.6	38.2	38.6	39.3	38.3	36.5	34.2
	10%	DB	51.9	49.8	50.8	52.5	54.9	57.0	56.6	56.4	55.1	54.4	51.9	51.0
		MCWB	33.6	34.3	35.6	36.8	38.7	38.8	38.8	39.7	40.4	38.5	36.4	34.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.5	44.1	44.8	45.7	47.4	48.3	48.3	49.3	49.1	48.4	46.5	44.7
		MCDB	51.3	49.6	51.5	51.8	54.3	54.6	52.5	53.4	53.7	52.6	51.5	49.9
	2%	WB	41.3	42.1	42.7	44.2	45.6	46.5	46.8	47.9	47.7	46.9	44.5	42.5
		MCDB	48.3	47.6	48.9	50.1	52.4	53.2	51.6	51.8	51.9	51.3	49.7	47.6
	5%	WB	39.3	40.5	41.3	42.8	44.3	45.2	46.0	46.8	46.6	45.7	43.0	41.0
		MCDB	46.2	46.5	47.2	48.8	50.9	52.3	51.1	51.1	51.2	50.4	48.7	46.2
	10%	WB	37.3	38.6	39.6	41.4	42.8	43.6	44.9	45.7	45.5	44.3	41.3	39.0
		MCDB	46.4	45.1	45.3	47.5	49.6	51.3	50.6	50.6	50.5	49.5	47.0	44.7
Mean Daily Temperature Range	5% DB	MDBR	14.2	13.4	13.8	14.3	14.5	14.8	14.6	14.5	13.7	13.9	13.4	13.0
		MCDBR	16.1	16.0	16.8	16.3	16.6	16.5	16.5	17.2	15.8	15.9	15.2	15.4
		MCWBR	12.4	13.9	15.0	15.4	15.6	14.2	14.1	14.7	15.5	14.3	13.9	12.3
	5% WB	MCDBR	14.2	13.6	14.5	13.7	14.1	13.9	13.2	13.1	13.1	13.2	13.5	12.3
		MCWBR	15.9	14.9	16.6	16.8	17.3	17.4	17.4	15.1	16.5	16.2	15.8	15.3
Clear-Sky Solar Irradiance	taub	0.228	0.230	0.236	0.250	0.249	0.243	0.244	0.241	0.240	0.239	0.235	0.228	
	taud	2.896	2.895	2.864	2.749	2.797	2.827	2.834	2.876	2.885	2.905	2.902	2.922	
	Ebn at Noon	343	344	342	334	330	329	329	333	336	337	337	340	
	Edh at Noon	21	23	24	27	26	25	25	24	23	22	21	20	
All-Sky Solar Radiation	RadAvg	1427	1526	1628	1761	1819	1894	1960	1914	1748	1570	1378	1297	
	RadStd	84	112	180	120	128	68	102	103	75	68	75	125	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+225	+225

Nomenclature: See separate page